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Date

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51.504 Machine Learning.

1) $f(x,y) = y - 3x$.

constraint = $x^2 + y^2 = 10$.

(a) $L(x,y,\lambda) = y - 3x + \lambda(x^2+y^2-10)$

(b) $\frac{\partial L}{\partial x} = 0$, $\frac{\partial L}{\partial y} = 0$, $x^2+y^2 = 10$ $\Rightarrow$ for extremum.

(i) $-3 + \lambda(2x) = 0$.
$\Rightarrow + \lambda(2x) = 3$.
$\Rightarrow x = +\frac{3}{2\lambda}$.

(ii) $1 + \lambda(2y) = 0$.
$y = -\frac{1}{2\lambda}$.

$\frac{9}{4\lambda^2} + \frac{1}{4\lambda^2} = 10$.
$\Rightarrow 10 = 10 \cdot 4\lambda^2$.
$\Rightarrow 10(1-4\lambda^2) = 0$.
$\Rightarrow 4\lambda^2 = 1$.
$\Rightarrow \lambda = \left(\pm\frac{1}{2}\right)$.

Therefore solutions = $\overset{\text{I}}{(-3,1)}$, $\overset{\text{II}}{(3,-1)}$.

$f(\text{I}) = 1 - 3(-3) = 10$.
$f(\text{II}) = -1 - 3(3) = -10 \rightarrow$ minimum.

$\therefore$ Min occurs @ $(3,-1)$.

(c) for inequality condition, we can impose $\lambda \geq 0$.

The basic form of the lagrangian remains the same.

$L(x,y,\lambda) = y - 3x + \lambda(x^2+y^2-10)$ $\boxed{\lambda \geq 0}$

Complementary slackness.

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(d) KKT / complementary slackness conditions:

(i) $-3 + 2\lambda x = 0$, $1 + 2\lambda y = 0$ [stationary].

(ii) Primary constraint: $x^2 + y^2 = 10$.

(iii) $\lambda \geq 0$!

(iv) $\lambda(x^2 + y^2 - 10) = 0$ $\Rightarrow$ either $\lambda = 0$ or $x^2 + y^2 - 10 = 0$.

If the constraint is $\boxed{\text{active}}$ $(x^2 + y^2 = 10)$ &

λ is positive, then $\lambda(x^2 + y^2 - 10) = 0 \Leftrightarrow x^2 + y^2 = 10$ [$\lambda \neq 0$]

OR

If constraint is $\boxed{\text{inactive}}$ $(x^2 + y^2 < 10)$ in that case $\lambda = 0$ which is NOT possible, so solution is @ constraint boundary

$\lambda = 0$ [inactive] ← Not possible

$\lambda \neq 0$ [active]. ← possible, only solution

2) (a) Linear separator equation: signum$(\theta_1 x + \theta_0)$

$\therefore$ $\theta_1 = 2$, $\theta_0 = -3$ $\Rightarrow$ signum$(2x - 3)$.

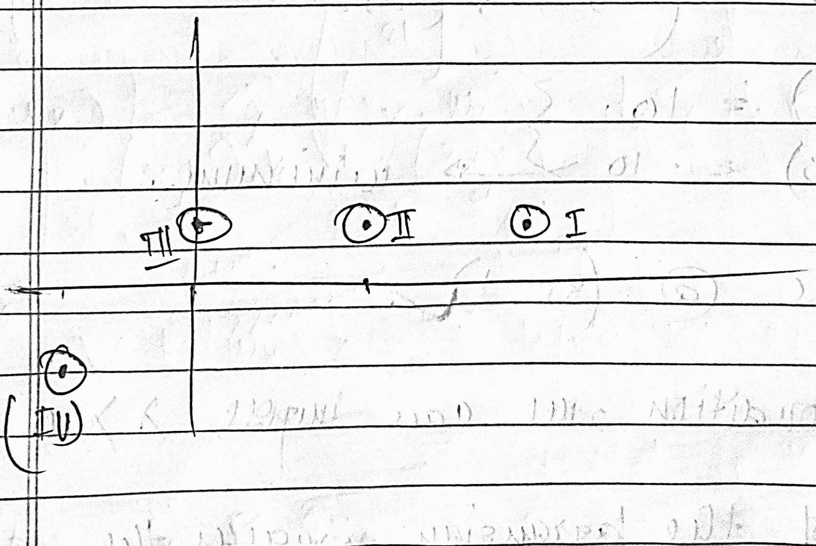
(b) x is input, y is output.

(I) $(2,1) \rightarrow 1$

(II) $(1,1) \rightarrow -1$

(III) $(0,1) \rightarrow -1$ → (misclassified)

(IV) $(-2,-1) \rightarrow -1$



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(c) $\theta_1' = \theta_1 + \alpha y^{(i)} x^{(i)}$

$\theta_0' = \theta_0 + \alpha y^{(i)}$,

choose $(1,1) \longrightarrow \theta_1' = \theta_1 + \alpha = 2+\alpha$.

$\theta_0' = \theta_0 + \alpha = -3+\alpha$

choose $(0,1) \longrightarrow \theta_1' = 2 + 0 = 2$.

$\left.\theta_0' = -3+\alpha = -3+\alpha.\right\}$

(d) Use same equation $\longrightarrow \theta_1 x + \theta_0 \Rightarrow \theta_1' x + \theta_0'$.

$\therefore$ choosing $(1,1) \longrightarrow (2+\alpha)x + (-3+\alpha) = 0$.

$h(1,1) > 1 \Rightarrow (2+\alpha) + (-3+\alpha) = -1+2\alpha > 0$.

$\therefore 2\alpha > 1$.

$\alpha > 1/2$.

$h(0,1) > 1 \Rightarrow (2+\alpha)\cdot 0 + (-3+\alpha) > 0$.

$\Rightarrow -3+\alpha > 0 \Rightarrow \boxed{\alpha > 3}$.

$\therefore \forall\, \alpha > 3$, for the perceptron updation algorithm we will have correct results. (ONE step correct)

choosing $(0,1) \longrightarrow 2x + (-3+\alpha)$.

$h(1,1) = 2 - 3 + \alpha > 0 \Rightarrow \alpha > 1$.

$h(0,1) = -3+\alpha > 0 \Rightarrow \alpha > 3$. $\Big\}\ \boxed{\alpha > 3}$

$\therefore$ Irrespective of which perceptron algorithm we choose the condition for $\boxed{\alpha > 3}$ for $\forall\, \alpha \in \mathbb{R}$.

For example $\boxed{\alpha = 4}$ is VALID.

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3.) (a) Kernel $= (xy+1)^2$.

pts $\Rightarrow$ $(1, +1)$, $(3, -1)$.

$x = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$. (sampled pts) | Target vector: $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

let test input x^*

cov : $[C]$. $\begin{bmatrix} K(1,1) & K(1,3) \\ K(3,1) & K(3,3) \end{bmatrix} = \begin{bmatrix} 4 & 16 \\ 16 & 100 \end{bmatrix}$.

$C^{-1} = \frac{1}{\|C\|} \begin{bmatrix} 100 & -16 \\ -16 & 4 \end{bmatrix} = \frac{1}{400-256} \begin{bmatrix} 100 & -16 \\ -16 & 4 \end{bmatrix}$.

$\therefore C^{-1} = \frac{1}{144} \begin{bmatrix} 100 & -16 \\ -16 & 4 \end{bmatrix}$.

New point : x^*

$K^{*T} = \begin{bmatrix} K(x^*, 1) & K(x^*, 3) \end{bmatrix} = \begin{bmatrix} (x+1)^2 & (3x+1)^2 \end{bmatrix}$.

$\therefore$ Mean calculation is given by :

$K^{*T} C^{-1} T = \frac{1}{144} \begin{bmatrix} (x+1)^2 & (3x+1)^2 \end{bmatrix} \begin{bmatrix} 100 & -16 \\ -16 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

$$m(x) = \frac{1}{144}\begin{bmatrix}(x+1)^2 & (3x+1)^2\end{bmatrix}\begin{bmatrix}100 & -16 \\ -16 & 4\end{bmatrix}\begin{bmatrix}1 \\ -1\end{bmatrix}.$$

$$= \frac{1}{144}\begin{bmatrix}(x+1)^2 & (3x+1)^2\end{bmatrix}\begin{bmatrix}116 \\ -20\end{bmatrix}.$$

$$= \frac{1}{144}\left[116(x+1)^2 - 20(3x+1)^2\right].$$

$$\boxed{m(x) = \frac{116}{144}(x+1)^2 - \frac{20}{144}(3x+1)^2.}$$

$$\boxed{m(x) = \frac{29}{36}(x+1)^2 - \frac{5}{36}(3x+1)^2.} = \frac{1}{9}\left[-4x^2 + 7x + 6\right] \text{ (ans.)}$$

Variance calculation.

$$\sigma^2 = \underbrace{k(x,x)}_{(i)} - \underbrace{k_*^T C^{-1} k_*}_{(ii)}.$$

(i) $k(x,x) = (x \cdot x + 1)^2 = (x^2+1)^2.$

(ii) $$k_*^T C^{-1} k_* = \frac{1}{144}\begin{bmatrix}(x+1)^2 & (3x+1)^2\end{bmatrix}\begin{bmatrix}100 & -16 \\ -16 & 4\end{bmatrix}\begin{bmatrix}(x+1)^2 \\ (3x+1)^2\end{bmatrix}.$$

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$$= \frac{1}{144}\begin{bmatrix}(x+1)^2 & (3x+1)^2\end{bmatrix}\begin{bmatrix}100(x+1)^2 - 16(3x+1)^2 \\ -16(x+1)^2 + 4(3x+1)^2\end{bmatrix}.$$

Let $(x+1)^2 = \alpha$, $(3x+1)^2 = \beta$.

$$\therefore K_x^T C^{-1} K_x = \frac{1}{144}\begin{bmatrix}\alpha & \beta\end{bmatrix}\begin{bmatrix}100\alpha - 16\beta \\ -16\alpha + 4\beta\end{bmatrix}.$$

$$= \frac{1}{144}\left[100\alpha^2 - 16\alpha\beta - 16\alpha\beta + 4\beta^2\right].$$

$$= \frac{\cancel{100}^{25}}{\cancel{144}_{36}}\alpha^2 - \frac{\cancel{32}^{8}}{\cancel{144}_{36}}\alpha\beta + \frac{\cancel{4}^{1}}{\cancel{144}_{36}}\beta^2.$$

$$\therefore \sigma^2 = (x^2+1)^2 - \left[\frac{25}{36}(x+1)^4 - \frac{8}{36}(x+1)^2(3x+1)^2 + \frac{1}{36}(3x+1)^4\right]$$

$$\boxed{\sigma^2(x) = \frac{1}{18}\left[x^4 - 8x^3 + 22x^2 - 24x + 9\right].}$$ (Ans.)

~~(b) $K(x,y) = xy - 5$. The condition for kernel, PSD i.e. $\|K(x,y)\|$ is $\geq 0 \ \forall x, y \in \mathbb{R}$.~~

3) $K(x, y) = xy - 5$.

Kernel function requirements.

1) eigenvectors are all ≥ 0.

2) For polynomial kernel, the function must be ≥ 0 for $\forall x_i \in R$.

If we choose $x = -5$, & $y = 2$

$K(-5, 2) = -10 - 5 = -15 < 0$

$\therefore K(-5, 2) < 0$ which is a violation of kernel requirements.

Therefore, $xy - 5$ is NOT a valid kernel.

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4) a) Data split in 80 : 20. Data in IRIS is quitably separated but ~~to be~~ on ~~test~~ same size we still stratify it.

b) we choose SVM and for kernel I chose gaussian. For γ [gaussian parameter we set it 'gamma = "scale"', random seed and set probability = 'False'.

~~In Question 1 (d) Point (IV)~~

~~KKT condition: $\Rightarrow \lambda(x^2 + y^2 - 10) = 0$~~

~~NOT < 0.~~